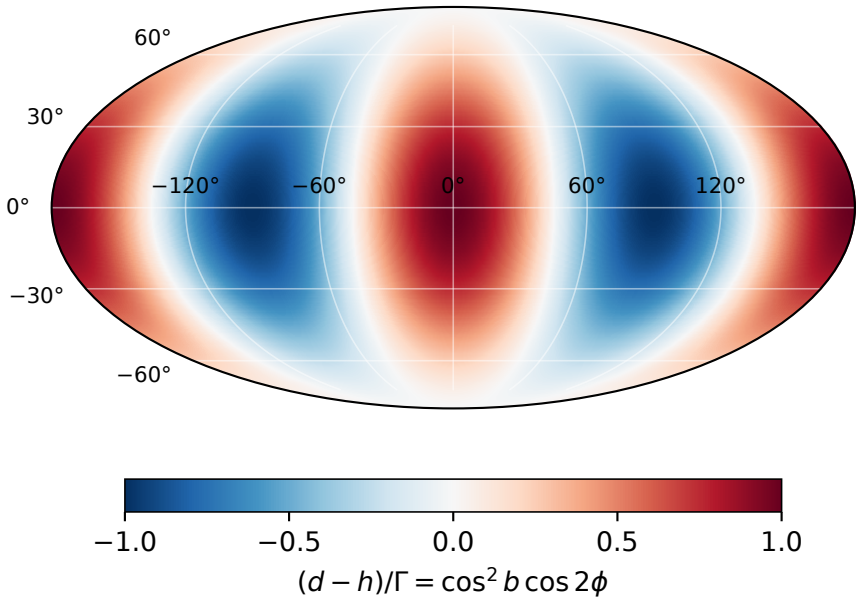
Local radial and transverse kinematics

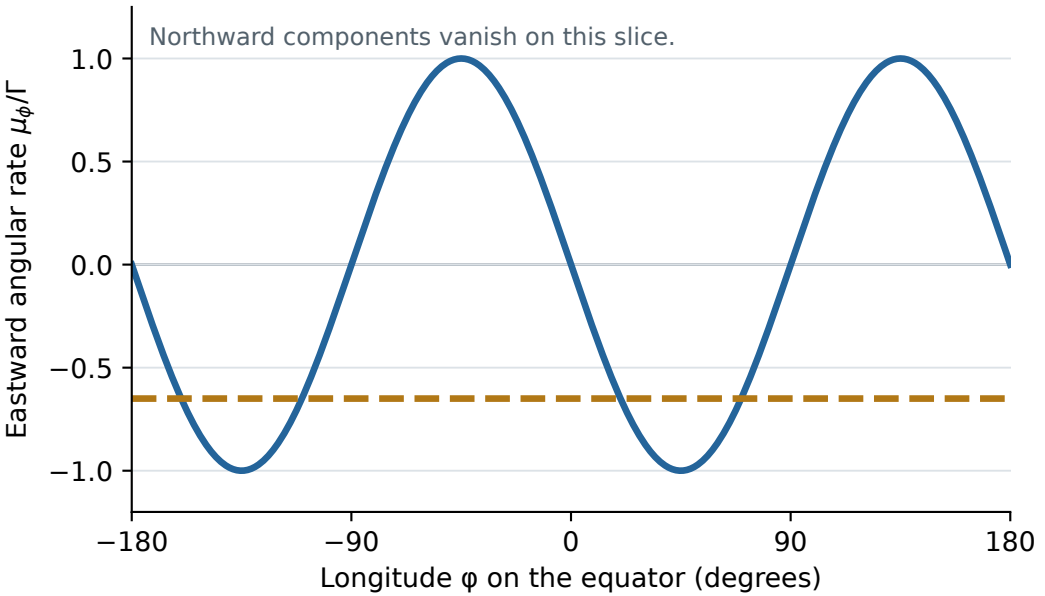
Analytic example • arbitrary rate unit $\Gamma > 0$ • no observational data

$h/\Gamma = 2, \quad S/\Gamma = \mathrm{diag}(1, -1, 0), \quad \alpha = 0, \quad v/\Gamma = (0, 0, 0.65), \quad \Omega = 0$

a Radial anisotropy over the full sky



b Exact transverse components on the equator



c Full-sphere linear response rank

Observation channels	Surviving coefficients	Rank / parameter count
Radial only	$h,\ S,\ \alpha$	9 / 12
Proper motion only	$S,\ v$	8 / 12
Joint, calibrated frame	$h,\ S,\ \alpha,\ v$	12 / 12
Joint, unknown triad spin	$h,\ S,\ \alpha,\ v + \Omega$	12 / 15

Ranks apply to the complete leading local response, not to the displayed equatorial slice.

Outward sky direction n; east is increasing φ. The B1 sign follows $\omega_{ij} = \varepsilon_{ijk}\,\omega_k$. An uncalibrated rotating triad measures v + Ω.